

Reproducibility and label leakage in machine learning for Q-CHAT-10 toddler autism screening: a synthetic-data study of a 1D CNN and classical classifiers

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Abstract

Background: Machine-learning (ML) studies on the public Q-CHAT-10 toddler autism screening dataset routinely report near-perfect accuracy. In that dataset, however, the class label is not a clinical diagnosis. It is generated by the questionnaire's own rule (item sum > 3), so the reported performance measures how well a model reproduces a threshold-defined screening label. **Methods:** We ran a methodological study on a fully specified synthetic surrogate (1054 records) with the dataset's schema, class balance and labelling rule. A one-dimensional convolutional neural network (1D CNN, 76,161 parameters) was compared with seven classical classifiers under a leakage-controlled pipeline. The pipeline drops the aggregate score and fits all encoders on training data only. Evaluation used 20 repeated stratified 80/20 splits; Wilson confidence intervals, exact McNemar tests and prevalence-adjusted predictive values on a held-out split; a feature-subset ablation; permutation importance; and 10 CNN re-initialisations. **Results:** Logistic regression reproduced the label without error on every split (accuracy 1.000 ± 0.000). The 1D CNN reached 0.997 ± 0.004 (AUC 1.000) and did not improve on it. Tree-based and probabilistic models were less accurate. With demographic and history attributes alone, all models were near chance (AUC ≤ 0.531). Even for the error-free model, the lower 95% bound of specificity (0.945 on 66 negatives) implies a positive predictive value as low as 0.354 at 3% population prevalence. **Conclusions:** On this benchmark, near-perfect ML performance shows that the scoring rule is learnable. It does not show diagnostic or screening validity, and a 1D CNN adds nothing over a linear model. We recommend that Q-CHAT-10 ML studies report the label's provenance, remove the score, compare against direct scoring, and validate against clinician-confirmed diagnoses. All numerical results here are from synthetic data.

Keywords: Autism spectrum disorder, Q-CHAT-10, Toddler screening, Label leakage, Reproducibility, Convolutional neural network, Synthetic data

1. Introduction

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition characterised by persistent differences in social communication and interaction, together with restricted and repetitive patterns of behaviour [1]. The most recent US surveillance estimated that 1 in 31 children aged eight years were identified with ASD in 2022 [2]. Early intensive behavioural intervention in toddlerhood improves cognitive, language and adaptive outcomes [3]. This makes accurate and scalable screening in the first three years important. Short parent-report instruments such as the M-CHAT-R/F [4] and the Q-CHAT-10 [5] are used as first-line screens.

Machine learning (ML) has been applied to autism instruments to shorten them, to fuse information across them, and to automate decisions [6, 7, 8, 9]. A widely used public resource is the toddler screening dataset collected with the ASDTests mobile application [10, 11]. On this and related datasets, many studies report accuracies close to 100% [14].

Such results need careful interpretation. In the toddler dataset, the class label "ASD traits" is not an independent clinical diagnosis. It is assigned when the Q-CHAT-10 item sum exceeds 3, which is the instrument's own referral rule [5]. A model given the ten items can therefore reproduce the label exactly. High accuracy on this benchmark measures agreement with the scoring rule, not the ability to identify autism. The dataset also contains the aggregate score column. Using it as a feature, or fitting encoders on the full dataset before splitting, leaks label information [15]. Throughout this paper we therefore use *threshold-defined screening label* for the prediction target and reserve "diagnosis" for clinician-confirmed status.

This paper is a methodological and reproducibility study. Because the original data file could not be obtained in our computing environment, all experiments were run on a synthetic surrogate that reproduces the dataset's schema, class balance and labelling rule. The generator is specified in full in Section 3.2. Its contributions are:

- a leakage-controlled, fully scripted pipeline in which the aggregate score is removed and all transformations are fitted on training data only;
- a comparison of a compact 1D CNN with seven classical classifiers over 20 repeated stratified splits, with con-

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fidence intervals, paired significance tests and prevalence-adjusted predictive values;

- a feature-subset ablation, permutation importance with uncertainty, and CNN re-initialisation experiments that show where the predictive information lies and how stable the results are;
- reporting recommendations for ML studies that use threshold-defined screening labels.

Section 2 reviews related work. Section 3 describes the data, the synthetic generator, the models and the protocol. Section 4 reports the results. Section 5 discusses them, and Section 6 concludes.

2. Related work

ML for shortening and automating autism instruments. Wall et al. [6] used alternating decision trees to reduce ADOS Module 1 to a few items with little loss of accuracy. Kosmicki et al. [7] used feature selection to find minimal behaviour sets for autism detection. Duda et al. [9] applied ML to distinguish autism from ADHD using the Social Responsiveness Scale. Bone et al. [8] examined ML-derived instruments critically. They showed how sampling choices and label definitions can make results look better than they are, and they proposed fusing information across instruments.

Q-CHAT-10 and the ASDTests datasets. Allison et al. [5] derived the Q-CHAT-10 from the full Q-CHAT. In their sample of 126 toddler cases and 754 controls, a cut-point of 3 gave a sensitivity of 0.91, a specificity of 0.89 and a positive predictive value (PPV) of 0.58. The published instrument advises considering referral when a child scores more than 3. Thabtah [10] discussed ML adaptation of ASD screening in relation to DSM-5, and later described the ASDTests application and its behavioural datasets [11]. Thabtah and Peebles [12] proposed a rule-induction model for autism detection. Both *Health Informatics Journal* articles [11, 12] are the subject of a 2021 expression of concern about their peer review [13]. We cite them only as the source and description of the dataset. Akter et al. [14] compared several classifiers and feature transformations on the ASDTests toddler, child, adolescent and adult datasets and reported very high accuracies.

Deep learning for ASD. Heinsfeld et al. [16] applied deep learning to resting-state fMRI connectivity from the ABIDE consortium. One-dimensional CNNs are well established for sequential and signal-like inputs [17]. On generic tabular data, however, deep models do not reliably outperform tree ensembles [18]. This motivates evaluating a CNN against strong classical baselines under an identical protocol.

3. Materials and methods

Fig. 1 summarises the workflow.

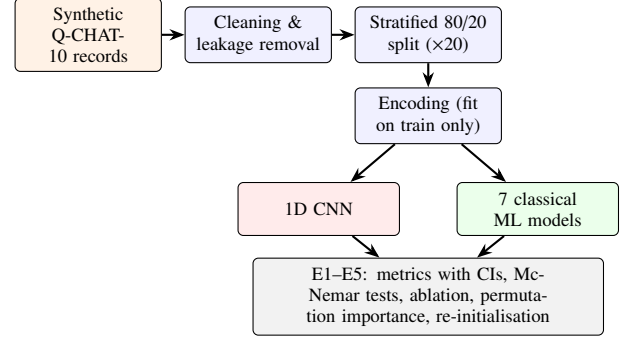


Figure 1: Overview of the experimental pipeline.

3.1. Reference dataset and label definition

The reference dataset is the public *Autism Screening Data for Toddlers* [11]. Its records were collected with the ASDTests mobile application from parents, carers and health professionals of toddlers aged 12–36 months, and they contain no direct identifiers. Each record holds the ten Q-CHAT-10 items (A1–A10), age in months, sex, ethnicity, history of neonatal jaundice, a family member with ASD, the respondent type, the Q-CHAT-10 score and the class label. The items are coded so that 1 denotes an answer consistent with an ASD trait. The class label is “ASD traits” when $\sum_{j=1}^{10} A_j > 3$ and “No ASD traits” otherwise. This is the instrument’s referral rule, not a clinical diagnosis. The descriptor articles [11] do not report clinical confirmation of the labels.

3.2. Synthetic surrogate

The original CSV could not be downloaded in the computing environment used for this study. We therefore generated a synthetic surrogate with the same size, schema, approximate class balance and labelling rule, using `asd.data.make_synthetic` with seed 42. **All numerical results in this paper were obtained on this surrogate, not on the public dataset.** The surrogate was generated once, and all splits, seeds and experiments use the same 1054 records. Table 1 lists every generative parameter. Each toddler i receives a latent status $z_i \sim \text{Bernoulli}(0.69)$. Each item has its own response probability, $p_{ij} \sim U(0.45, 0.85)$ if $z_i = 1$ and $p_{ij} \sim U(0.03, 0.30)$ otherwise, and $A_{ij} \sim \text{Bernoulli}(p_{ij})$. Items are therefore independent given z_i and p_{ij} , and they are correlated with each other only through z_i . Sex, jaundice and family history depend on z_i with the small effect sizes in Table 1. Age, ethnicity and respondent type are independent of everything else. The label is computed from the items with the published rule, *not* from z_i . The label therefore differs from the latent status for some toddlers, and demographic attributes relate to the label only indirectly through z_i .

The surrogate matches the structural property that drives our conclusions: the label is a deterministic function of the items. It does not reproduce the true joint distribution of the public data, including real inter-item correlations, real demographic associations and data-entry artefacts. Absolute numbers, especially

Table 1: Generative parameters of the synthetic surrogate ($n = 1054$, seed 42). z : latent status. “Indep.”: independent of z .

Variable	If $z = 1$	If $z = 0$
Latent status z	Bernoulli(0.69)	
Item prob. p_{ij} (per item)	$U(0.45, 0.85)$	$U(0.03, 0.30)$
Items A_{ij}	Bernoulli(p_{ij}), independent given p_{ij}	
Sex (male)	0.74	0.60
Neonatal jaundice	0.30	0.22
Family member with ASD	0.18	0.12
Age (months)	Indep.; uniform integer 12–36	
Ethnicity (11 categories)	Indep.; uniform	
Respondent (5 categories)	Indep.; (0.93, 0.03, 0.01, 0.02, 0.01)	
Label	“ASD traits” iff $\sum_j A_{ij} > 3$	

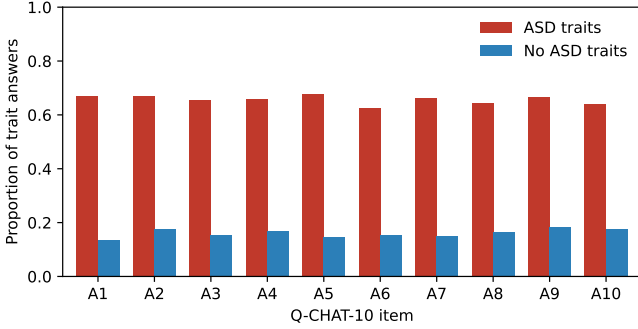


Figure 2: Proportion of trait-indicating answers to each Q-CHAT-10 item by label class (synthetic surrogate).

those for demographic features, reflect the generator’s parameter choices. They must be re-estimated on the public data, which the released script does with a single command.

The surrogate contains 1054 toddlers: 722 (68.5%) with the ASD-traits label and 332 without. 72.6% are male, mean age is 23.9 ± 7.1 months, 28.2% have a history of jaundice and 15.8% report a family member with ASD. Fig. 2 shows the proportion of trait answers per item in each class. Fig. 3 shows the score distribution, where the classes separate exactly at the threshold.

3.3. Preprocessing and leakage control

The case identifier is removed. The Q-CHAT-10 score is also removed, because it is the sum of A1–A10 and defines the label directly. With the score included, a single threshold on that one column classifies every record correctly. Categorical strings are normalised (trimmed and lower-cased). Sex, jaundice and family history are coded as binary variables. Ethnicity and respondent type are one-hot encoded, and categories unseen in training are mapped to all zeros. Age is min–max scaled to $[0, 1]$, and the binary items pass through unchanged. The encoded vector has $d = 29$ features. Each split is stratified by class into 80% training and 20% test data. The encoder and scaler are fitted on the training portion only.

3.4. Classical machine-learning models

Seven scikit-learn [19] classifiers serve as baselines:

- logistic regression (LR), ℓ_2 -regularised with $C=1$;

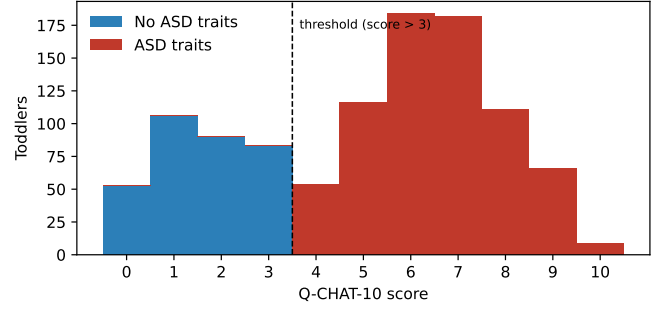


Figure 3: Distribution of the Q-CHAT-10 score (synthetic surrogate). The dashed line marks the rule (score > 3) that defines the label, and the two classes do not overlap across it.

Table 2: Layer-by-layer structure of the 1D CNN, taken from the executable model (`model.summary()`).

Layer (configuration)	Output shape	Parameters
Input	29×1	0
Conv1D (32, $k=3$, ReLU)	29×32	128
Conv1D (64, $k=3$, ReLU)	29×64	6,208
MaxPool1D (size 2)	14×64	0
Dropout (0.25)	14×64	0
Conv1D (64, $k=3$, ReLU)	14×64	12,352
Flatten	896	0
Dense (64, relu)	64	57,408
Dropout (0.3)	64	0
Dense (1, sigmoid)	1	65
Total (all trainable)		76,161

- an RBF-kernel support vector machine (SVM, $C=1$) [20], with Platt scaling [21] fitted by internal cross-validation;
- k -nearest neighbours (KNN, $k=5$);
- a decision tree (DT) with a maximum depth of 6;
- a random forest (RF) [22] with 300 trees;
- gradient boosting (GB) [23] with 100 trees of depth 3;
- Gaussian naive Bayes (NB).

Hyper-parameters were fixed a priori and not tuned. Model-internal randomness uses the split’s seed.

3.5. One-dimensional CNN

The encoded record $\mathbf{x} \in \mathbb{R}^d$ is treated as a one-channel sequence of length d . Because A1–A10 occupy adjacent positions, the convolution kernels can combine neighbouring items, and weight sharing keeps the parameter count small [17]. The architecture is shown in Fig. 4 and Table 2; the table is generated from the executable model. Flattening rather than global pooling keeps each feature’s position, which is informative because each position is a specific question. Max-pooling uses floor rounding, so the sequence length goes from 29 to 14, and the model has 76,161 trainable parameters.

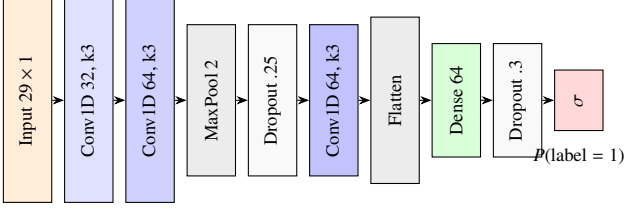


Figure 4: Architecture of the 1D CNN. ReLU follows every convolution and the hidden dense layer.

The CNN is trained with Adam [25] (learning rate 10^{-3} , batch size 32) to minimise binary cross-entropy,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (1)$$

A class-stratified 15% of each training set is held out for validation. The validation subset is drawn with the run’s seed, so it differs between runs. Dropout [24] is applied as in Table 2. The learning rate is halved after 5 epochs without improvement in validation loss. Training stops after 20 such epochs (maximum 150), and the weights with the lowest validation loss are restored. Weights use the Keras defaults (Glorot-uniform kernels, zero biases). Python, NumPy and TensorFlow seeds are set per run, and TensorFlow’s deterministic-operation mode is enabled [26]. The CNN was trained 20 times in E2 (once per split), 3×20 times in E3 and 10 further times in E5.

3.6. Experimental protocol

Five experiments were run.

- **E1 (held-out split):** one fixed split (seed 42; 211 test records, 145 positive and 66 negative). It provides the confusion matrices, ROC curves, screening metrics with Wilson 95% confidence intervals [27], and exact McNemar tests [28] comparing each model with LR on the same test records.
- **E2 (repeated splits):** all eight models on 20 independent stratified splits (seeds 0–19). We report the mean, standard deviation and the 2.5th–97.5th percentile range across splits. Together the test sets cover 1045 of the 1054 records. The splits overlap, so they are not independent samples, and the SD describes split-to-split variability rather than a standard error.
- **E3 (ablation):** LR, RF and the CNN trained on (i) the ten items only, (ii) demographic and history attributes only, and (iii) all features, using the same 20 splits.
- **E4 (importance):** permutation importance on the test sets of the first 5 E2 splits, with 30 permutations per variable and split. Importance is the mean decrease in ROC-AUC, and one-hot columns of a variable are permuted together. We also report the LR coefficients from E1.
- **E5 (CNN stability):** the CNN retrained 10 times on the E1 split with different initialisation and validation seeds.

Metrics. With TP, TN, FP and FN the counts of true and false positives and negatives (positive = ASD-traits label), we report

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{Sensitivity} = \frac{TP}{TP+FN}, \quad (2)$$

$$\text{Specificity} = \frac{TN}{TN+FP}, \quad \text{PPV} = \frac{TP}{TP+FP}, \quad \text{NPV} = \frac{TN}{TN+FN}, \quad (3)$$

$$\text{LR}^+ = \frac{\text{Sens}}{1-\text{Spec}}, \quad \text{LR}^- = \frac{1-\text{Sens}}{\text{Spec}}, \quad (4)$$

together with the F1-score and ROC-AUC, all at a 0.5 threshold. PPV and NPV depend on prevalence, which is 68.5% in the data. We therefore also report the PPV at a population prevalence of $\pi = 0.03$, computed as $\text{Sens} \pi / [\text{Sens} \pi + (1 - \text{Spec})(1 - \pi)]$ from both the point estimates and the lower confidence bounds. We did not use DeLong’s test [29] because several AUCs equal 1.000, where the test is uninformative. Paired comparisons use McNemar’s test on the E1 predictions.

Computational environment. Experiments ran on CPU only (Intel Xeon Processor @ 2.10GHz, 4 cores; Linux 6.18.44) with Python 3.11.15, TensorFlow 2.21.0, scikit-learn 1.9.1, NumPy 2.4.6 and pandas 3.0.6. The command `python experiments.py --synthetic` regenerates every table, figure and number in this paper.

4. Results

All results in this section are on the synthetic surrogate.

4.1. Comparison over repeated splits (E2)

Table 3 summarises E2. LR reproduced the threshold-defined label without error on every split (accuracy 1.000 ± 0.000). The 1D CNN reached 0.997 ± 0.004 (sensitivity 0.997, specificity 0.997, AUC 1.000), close to but not above LR. The RBF SVM followed. The tree ensembles kept sensitivity high but lost specificity, because axis-aligned splits approximate a threshold on a sum of ten binary variables only coarsely. Naive Bayes, whose independence assumption does not hold for the items given the label, also lost specificity. The single decision tree was weakest (accuracy 0.895).

4.2. Held-out screening metrics and paired tests (E1)

Table 4 and Figs. 5–6 report E1. On this split the CNN’s McNemar p -value against LR is 0.500. The confidence intervals are wide despite point estimates of 1.000, because the test set contains only 66 negatives. For LR, the lower Wilson bounds are 0.974 for sensitivity and 0.945 for specificity. At a 3% population prevalence, a specificity at that lower bound would give a PPV of only 0.354. The data therefore cannot rule out that most positive screens would be false positives in a general population, even for a model that makes no errors here. For the CNN, the corresponding value is 0.348.

Table 3: Test-set performance over 20 repeated stratified 80/20 splits of the synthetic dataset: mean $\pm$ SD, and the 2.5th–97.5th percentile range of accuracy across splits. Rows are sorted by F1-score.

Model	Accuracy	Acc. range	Sensitivity	Specificity	F1	AUC
Logistic Regression	1.000 $\pm$ 0.000	1.000–1.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
1D CNN	0.997 $\pm$ 0.004	0.991–1.000	0.997 $\pm$ 0.005	0.997 $\pm$ 0.006	0.998 $\pm$ 0.003	1.000 $\pm$ 0.000
SVM (RBF)	0.994 $\pm$ 0.006	0.981–1.000	0.995 $\pm$ 0.007	0.992 $\pm$ 0.010	0.996 $\pm$ 0.005	1.000 $\pm$ 0.000
Gradient Boosting	0.983 $\pm$ 0.011	0.967–1.000	0.990 $\pm$ 0.009	0.967 $\pm$ 0.028	0.988 $\pm$ 0.008	0.999 $\pm$ 0.001
Random Forest	0.980 $\pm$ 0.009	0.960–0.991	0.998 $\pm$ 0.004	0.941 $\pm$ 0.028	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
KNN	0.966 $\pm$ 0.016	0.936–0.991	0.961 $\pm$ 0.019	0.976 $\pm$ 0.024	0.975 $\pm$ 0.012	0.996 $\pm$ 0.003
Naive Bayes	0.951 $\pm$ 0.013	0.926–0.969	0.983 $\pm$ 0.012	0.882 $\pm$ 0.033	0.965 $\pm$ 0.009	0.981 $\pm$ 0.016
Decision Tree	0.895 $\pm$ 0.021	0.857–0.934	0.914 $\pm$ 0.020	0.853 $\pm$ 0.047	0.923 $\pm$ 0.016	0.922 $\pm$ 0.024

Table 4: Screening metrics on the E1 test split ($n = 211$: 145 positive, 66 negative) with Wilson 95% confidence intervals. PPV and NPV are at the dataset’s prevalence; PPV_{3%} re-weights the same sensitivity and specificity to a 3% population prevalence (point estimate / value at the lower CI bounds). p : exact McNemar test against logistic regression.

Model	Sensitivity (95% CI)	Specificity (95% CI)	PPV	NPV	LR+	LR–	PPV _{3%}	p
Logistic Regression	1.000 (0.974–1.000)	1.000 (0.945–1.000)	1.000	1.000	∞	0.000	1.000 / 0.354	–
SVM (RBF)	1.000 (0.974–1.000)	1.000 (0.945–1.000)	1.000	1.000	∞	0.000	1.000 / 0.354	1.000
1D CNN	0.986 (0.951–0.996)	1.000 (0.945–1.000)	1.000	0.971	∞	0.014	1.000 / 0.348	0.500
Gradient Boosting	1.000 (0.974–1.000)	0.924 (0.835–0.967)	0.967	1.000	13.2	0.000	0.290 / 0.154	0.062
Random Forest	1.000 (0.974–1.000)	0.894 (0.797–0.948)	0.954	1.000	9.4	0.000	0.226 / 0.129	0.016
KNN	0.979 (0.941–0.993)	0.924 (0.835–0.967)	0.966	0.953	12.9	0.022	0.286 / 0.150	0.008
Naive Bayes	0.986 (0.951–0.996)	0.773 (0.658–0.857)	0.905	0.962	4.3	0.018	0.118 / 0.079	<0.001
Decision Tree	0.924 (0.869–0.957)	0.742 (0.626–0.833)	0.887	0.817	3.6	0.102	0.100 / 0.067	<0.001

Table 5: Feature-subset ablation (mean $\pm$ SD over 20 splits).

Model	Accuracy	F1	AUC
<i>Q-CHAT-10 items only</i>			
Logistic Regression	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Random Forest	0.981 $\pm$ 0.010	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
1D CNN	1.000 $\pm$ 0.001	1.000 $\pm$ 0.001	1.000 $\pm$ 0.000
<i>Demographics only</i>			
Logistic Regression	0.674 $\pm$ 0.013	0.804 $\pm$ 0.010	0.531 $\pm$ 0.040
Random Forest	0.597 $\pm$ 0.030	0.717 $\pm$ 0.025	0.515 $\pm$ 0.043
1D CNN	0.686 $\pm$ 0.004	0.814 $\pm$ 0.002	0.530 $\pm$ 0.042
<i>All features</i>			
Logistic Regression	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Random Forest	0.980 $\pm$ 0.009	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
1D CNN	0.997 $\pm$ 0.004	0.998 $\pm$ 0.003	1.000 $\pm$ 0.000

4.3. CNN stability across initialisations (E5)

Across 10 re-initialisations on the E1 split, CNN test accuracy was 1.000 ± 0.000 (minimum 1.000). The E1 run itself (seed 42) made 2 errors on the same test set (Table 4), so which records the CNN misclassifies depends on its initialisation and validation subset. Early stopping ended training between epochs 49 and 119. Fig. 7 shows the mean $\pm$ SD learning curves. Validation loss fell sharply within the first few epochs and then plateaued, and training and validation curves stayed close.

4.4. Feature-subset ablation (E3)

Table 5 shows that the items carry the predictive signal for this label. With the items alone, LR reached 1.000 and the CNN 1.000, matching the full-feature models. With only age, sex, ethnicity, jaundice, family history and respondent type, all three models were near chance (AUC: LR 0.531, RF 0.515, CNN

0.530). In this setting the CNN’s accuracy (0.686) is close to the proportion of the majority class, because it predicted the ASD-traits label for nearly every test record. Its F1-score is therefore inflated by class imbalance, which is why accuracy and F1 must be read together with AUC and specificity. Under the generator, demographic attributes depend on the latent status only weakly and on the label only indirectly (Table 1). This result therefore says nothing about the clinical value of these attributes.

4.5. Feature importance (E4)

Fig. 8a shows permutation importance with error bars (SD over 5 splits $\times$ 30 permutations). For all three models, every item has a positive importance. For the CNN it ranges from 0.004 to 0.012 (largest for A1). The largest absolute importance of any demographic or history variable, for any model, is <0.001. The LR coefficients (Fig. 8b) tell the same story. The item coefficients all lie between 1.93 and 2.81, close to the equal weights of the scoring rule, while no demographic coefficient exceeds 0.46 in magnitude. These rankings describe how the models reproduce an additive rule on synthetic data. They should not be read as a clinical ordering of the items.

5. Discussion

What near-perfect accuracy means here. The reference label is a deterministic function of the inputs, $y = \mathbf{1}[\sum_j A_j > 3]$. A linear model represents this function exactly, so LR reaches 100%, and the CNN approximates it closely. On this label, a learned model has no practical advantage over adding up the ten answers. Reported accuracy gains on the benchmark are gains in reproducing a published scoring rule. They are not gains in detecting autism. This conclusion follows from how the label is

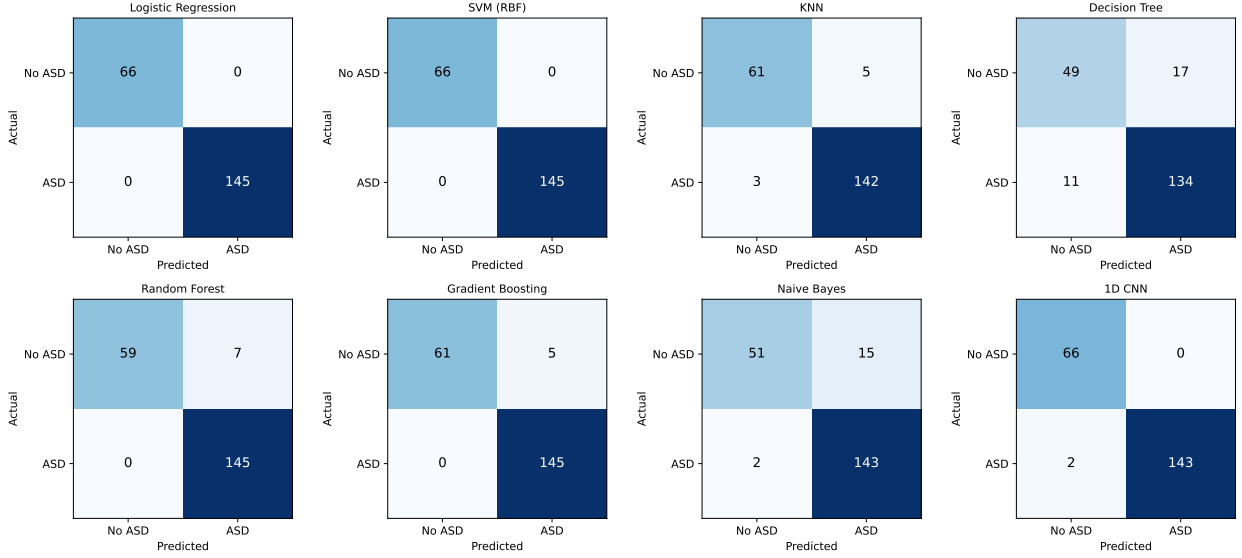


Figure 5: Confusion matrices on the E1 test split (seed 42; 211 records).

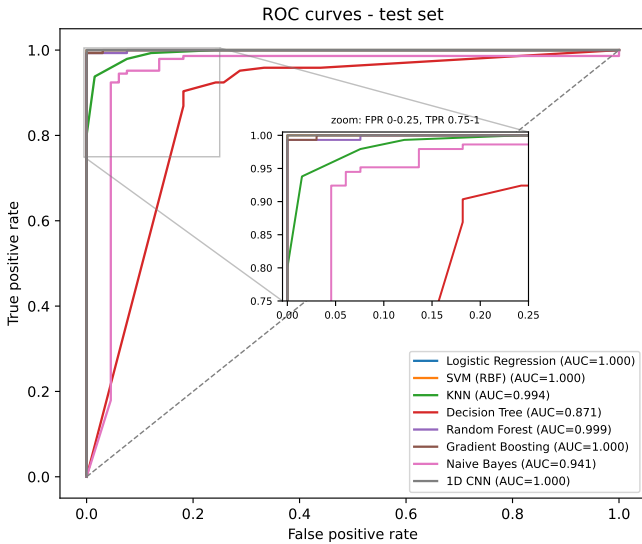


Figure 6: ROC curves on the E1 test split. The inset magnifies the region $\text{FPR} \leq 0.25$, $\text{TPR} \geq 0.75$, where the curves differ.

defined, not from the synthetic data. It therefore applies to the public dataset, and to any other dataset whose label is computed from the model’s inputs.

Role of the CNN. The CNN’s shared kernels over adjacent items and its dense head can approximate a weighted item sum. Dropout and early stopping keep it stable across splits and initialisations. However, LR matches its performance with about 30 parameters instead of 76,161, trains in milliseconds, and exposes coefficients that can be checked directly against the scoring rule. A CNN would be worth revisiting for richer inputs, such as item response times, free-text answers or video, and with independent clinical labels.

Leakage and reporting. With the score column retained, one threshold on that column classifies every record correctly. We recommend that ML studies on questionnaire-derived labels (i) state how the label was generated, (ii) list the input columns and confirm that label-defining aggregates are excluded, (iii) state where preprocessing was fitted, (iv) compare against direct scoring of the instrument, and (v) report confidence intervals and prevalence-adjusted predictive values rather than accuracy alone.

Towards clinically meaningful evaluation. A useful ML screen must be validated against an independent diagnostic reference, such as ADOS-2 or multidisciplinary clinical consensus, in a population with realistic prevalence. The label prevalence here (68.5%) is far above the roughly 3% seen in the general population [2]. As Table 4 shows, PPV at population prevalence is sensitive to small changes in specificity. False positives and false negatives both carry costs. Missed cases delay intervention, while false positives add family burden and demand on assessment services. Operating points should therefore be chosen for the intended setting. Future work should use clinician-confirmed outcomes, report calibration, and test performance across age, sex and ethnic groups. Instrument fusion [8] and item reduction [6, 7] are more promising uses of ML than re-learning a published threshold.

Limitations. First and most important, all numerical results come from a synthetic surrogate (Section 3.2). The generator reproduces the labelling rule and class balance but assumes items that are independent given latent status, and demographic effect sizes chosen by us. Results for demographic features and non-linear models in particular may differ on the public data. The qualitative conclusion about the label follows from its definition and does not depend on this choice. Second, the public dataset itself is self-selected through a mobile application, has

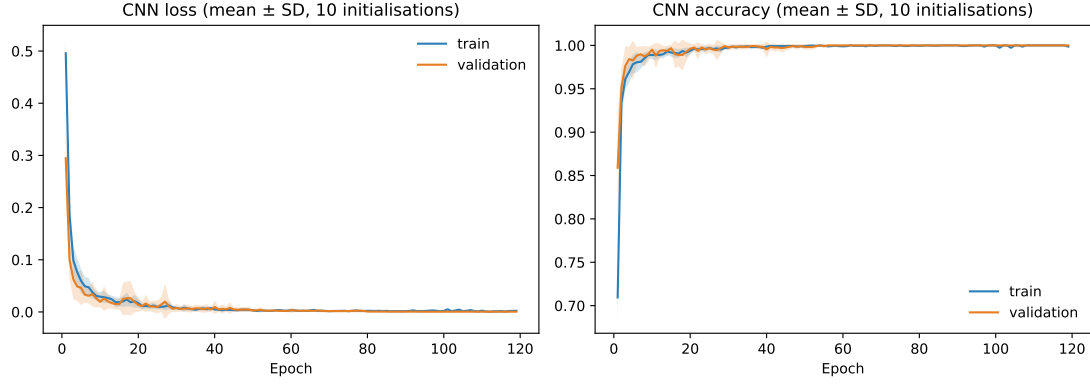


Figure 7: CNN learning curves on the E1 split: mean $\pm$ SD over 10 initialisations. Runs end at different epochs, so later epochs average fewer runs.

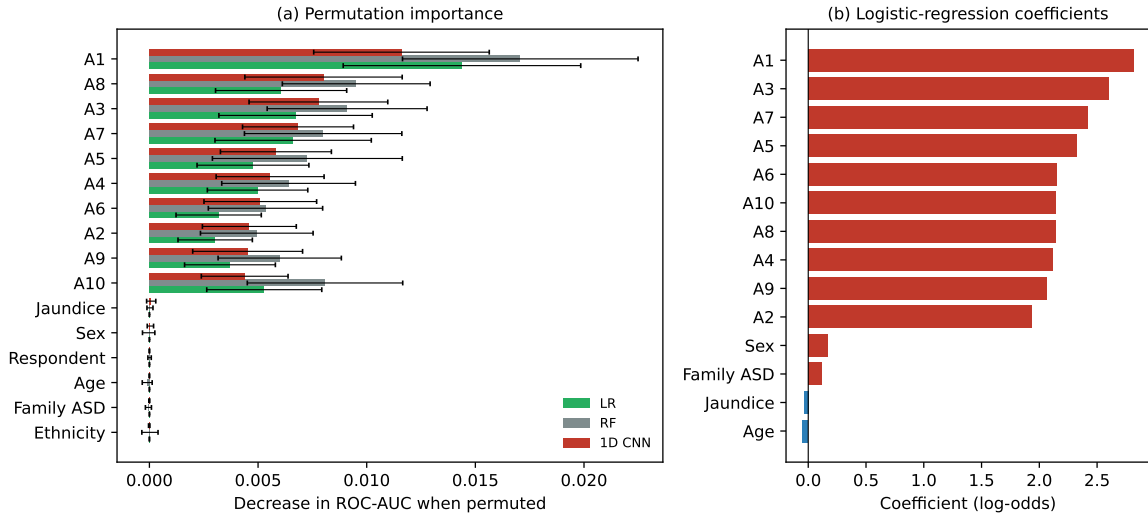


Figure 8: (a) Permutation importance (mean decrease in test ROC-AUC $\pm$ SD over 5 splits $\times$ 30 permutations). (b) Logistic-regression coefficients on the E1 split (items, binary history variables and scaled age).

no clinical confirmation of diagnosis, and its descriptor articles are subject to an expression of concern [13]. Third, hyperparameters were not tuned. This avoids optimistic bias, but it may disadvantage some models on harder labels. Fourth, the E2 splits overlap, and the E1 test set is small (66 negatives), which limits the precision of specificity estimates.

6. Conclusion

On a synthetic surrogate that reproduces the Q-CHAT-10 toddler dataset’s schema and labelling rule, a leakage-controlled comparison showed the following. Logistic regression reproduces the threshold-defined screening label exactly, and a 1D CNN comes close but does not improve on it. All of the predictive information lies in the ten items. Even an error-free model leaves wide uncertainty about predictive value at population prevalence. Near-perfect accuracy on this benchmark therefore shows that the scoring rule is learnable, not that a model is clinically valid. Re-running the released pipeline on the public data,

and above all evaluating models against clinician-confirmed diagnoses, are the necessary next steps.

CRedit authorship contribution statement

Subrahmanyam Tanala: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Code (Anthropic) to draft software, run experiments and draft the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and code availability

The public toddler screening dataset is described in [11]. No data from it are included in, or were used for, the results of this paper. The synthetic surrogate, the generator, the model code and the scripts that produce every table and figure are available at <https://github.com/subrahmanyamtanala-rgb/-autism-spectrum-disorders-classification-model>.

Ethics statement

This study used only synthetic data generated by the authors' code, so no human participants were involved and no ethical approval was required. The public dataset that the surrogate imitates is distributed as anonymised records without direct identifiers. Its descriptor articles [11] do not describe the consent or ethical-approval procedures of the original data collection, and those articles are subject to an expression of concern [13].

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